Medical Microbiology

Lecture Ten

Staphylococcus

Third stage- College of Dentistry

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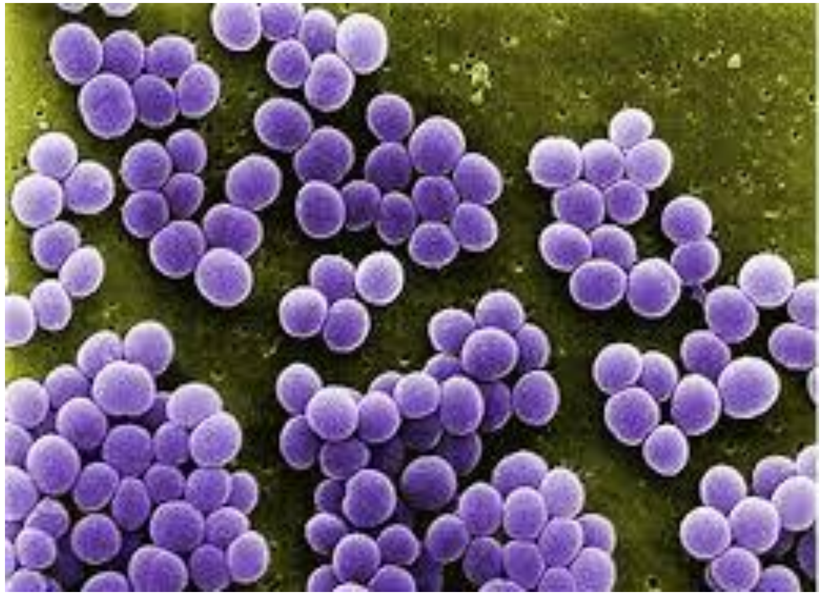
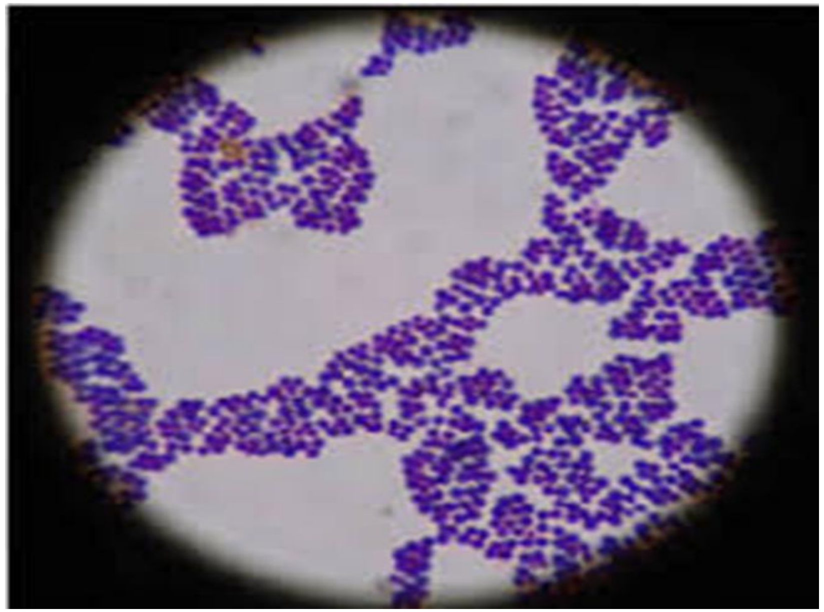
Staphylococcus and Staphylococcus aureus:

- **Staphylococcus** is a genus of Gram-positive, facultative anaerobic bacteria that appear as **grape-like clusters** under the microscope.
- These bacteria are widely distributed in the environment and are commonly found on human skin and mucous membranes.
- Some species are harmless, while others can cause serious infections.
- One of the most clinically significant species is *Staphylococcus aureus*, which is a major human pathogen.

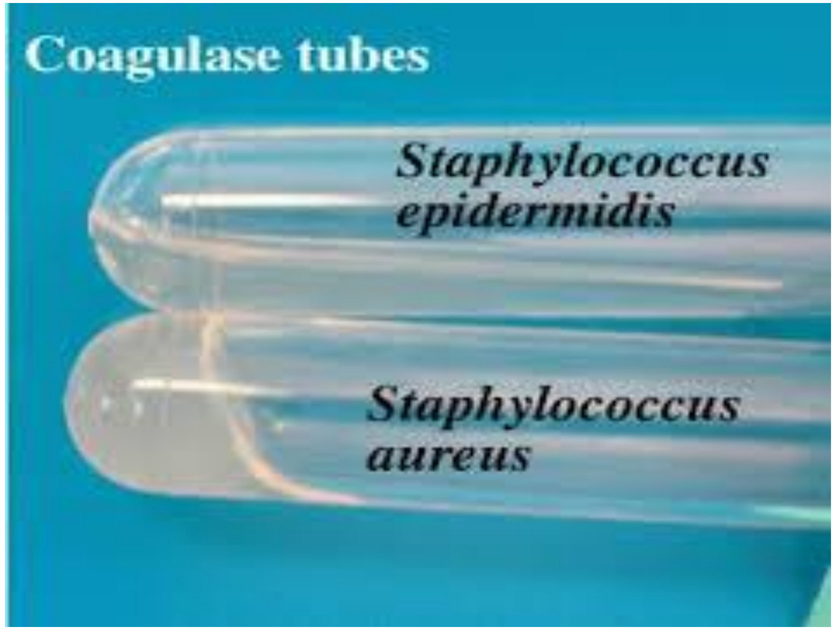
Characteristics of *Staphylococcus aureus* :





- **Gram-positive cocci** (retains the crystal violet stain) arranged in clusters.
- **Metabolism: Facultative anaerobe** (can survive with or without oxygen).
- **Catalase-positive** (distinguishes it from Streptococcus species).

- **Coagulase-positive** (distinguishes it from other Staphylococcus species like S. epidermidis).
- **Produces golden-yellow colonies** on culture media.
- **Beta-hemolytic** on blood agar (complete hemolysis).



Virulence Factors

Staphylococcus aureus has multiple virulence factors that allow it to cause a wide range of diseases:

1. Structural Components

- **Protein A**: Clumping Factors , preventing opsonization and phagocytosis.
- **Teichoic acids**: Help with adhesion to surfaces.
- **Capsule**: Inhibits phagocytosis.

2. Enzymes

- **Coagulase**: Converts fibrinogen to fibrin, forming a clot that protects the bacteria from the immune system.
- **Hyaluronidase**: Breaks down connective tissue, facilitating spread.
- **Staphylokinase**: Dissolves clots to help bacterial dissemination.
- **Lipases & Proteases**: Assist in tissue invasion and survival.

3. Toxins

- **Hemolysins:** Destroy red blood cells, Tissue Damage and immune evasion.
- **Leukocidins**: Kill white blood cells.
- **Enterotoxins**: Cause food poisoning.
- **Exfoliative toxin**: Leads to **staphylococcal scalded skin syndrome (SSSS)**.
- **Toxic Shock Syndrome Toxin (TSST-1)**: Causes **toxic shock syndrome**.

4. Biofilm formation

Polysaccharide Intercellular Adhesion (PIA): Allows colonization and persistence on dental materials, damaged tissue, and within abscesses, significantly increasing antibiotic resistance.

Diseases Caused by Staphylococcus aureus

Staphylococcus aureus can cause a variety of infections, ranging from mild skin infections to life-threatening diseases:

1. Skin Infections

- Folliculitis.
- Furuncles (boils).
- Cellulitis.

2. Deep Infections

- **Osteomyelitis** (bone infection).
- **Septic arthritis** (joint infection).
- **Endocarditis** (infection of heart valves).
- **Pneumonia.**

3. Toxin-Mediated Diseases

- **Food poisoning** (due to enterotoxins).
- **Toxic Shock Syndrome (TSS).**
- **Staphylococcal Scalded Skin Syndrome (SSSS).**

Types of Staphylococcus Associated with Dental Caries and Oral Problems

1. *Staphylococcus aureus*

- It may be present in the mouth and contribute to **gingival infections or dental abscesses**, but it is not the primary cause of tooth decay.
- It can cause **infections after oral surgeries**, such as **dental implant procedures**.

2. *Staphylococcus epidermidis*

- Naturally found in the oral cavity but **rarely causes tooth decay**.
- It can cause **gum or root infections**, especially in people with **dental prosthetics**, such as crowns and bridges.

3. *Staphylococcus warneri & Staphylococcus hominis*

- They have been found in some cases of **oral and dental infections**, but they are **not primary causes of tooth decay**.

Staph. aureus Association with Oral and Maxillofacial Infections:

- **Endodontic and Periapical Infections:** Although not the most common pathogen, *S. aureus* has been detected in persistent endodontic infections and dental abscesses, representing approximately **0.7%–15%** of cases.
- **Orofacial Soft Tissue Infections:** *S. aureus* is linked to several soft tissue conditions, including **angular cheilitis**, **facial cellulitis**, and various **mucosal lesions**.

- **Periodontal and Peri-implant Diseases:**
Growing evidence indicates that *S. aureus* may worsen or maintain **periodontitis** and **peri-implantitis**, largely due to its strong **biofilm-forming ability** and interactions with other oral pathogens.
- **Systemic Risk:**
Oral colonization by *S. aureus* can serve as a source for **systemic infections**, such as **infective endocarditis**. This risk increases in **susceptible dental patients**, especially following **invasive dental procedures** that can induce bacteremia.

Q/How Can Staphylococcus Affect Teeth?

- May contribute to **secondary infections** in the mouth, such as **dental abscesses**.
- Can lead to **post-treatment infections**, especially in **immunocompromised patients**.
- Some recent studies suggest the presence of **Staphylococcus aureus** in **dental plaque**, which may play a role in worsening tooth decay.

Lab. Diagnosis:

-Grow on ordinary media.

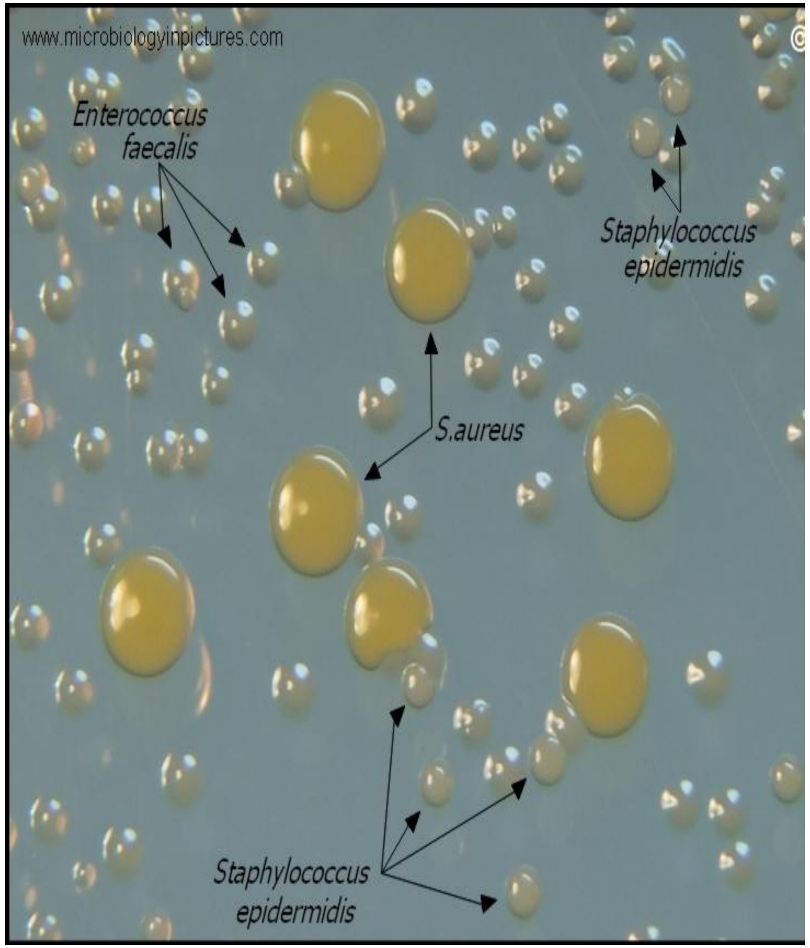
-Temp 10-42 C. optimum is 37 C.

-PH range 7.4 - 7.6 .

-Aerobes Or Facultative anaerobes.

-On nutrient agar:-

-Yellow pigment colony and easily emulsifiable, shiny , smooth circular Large.



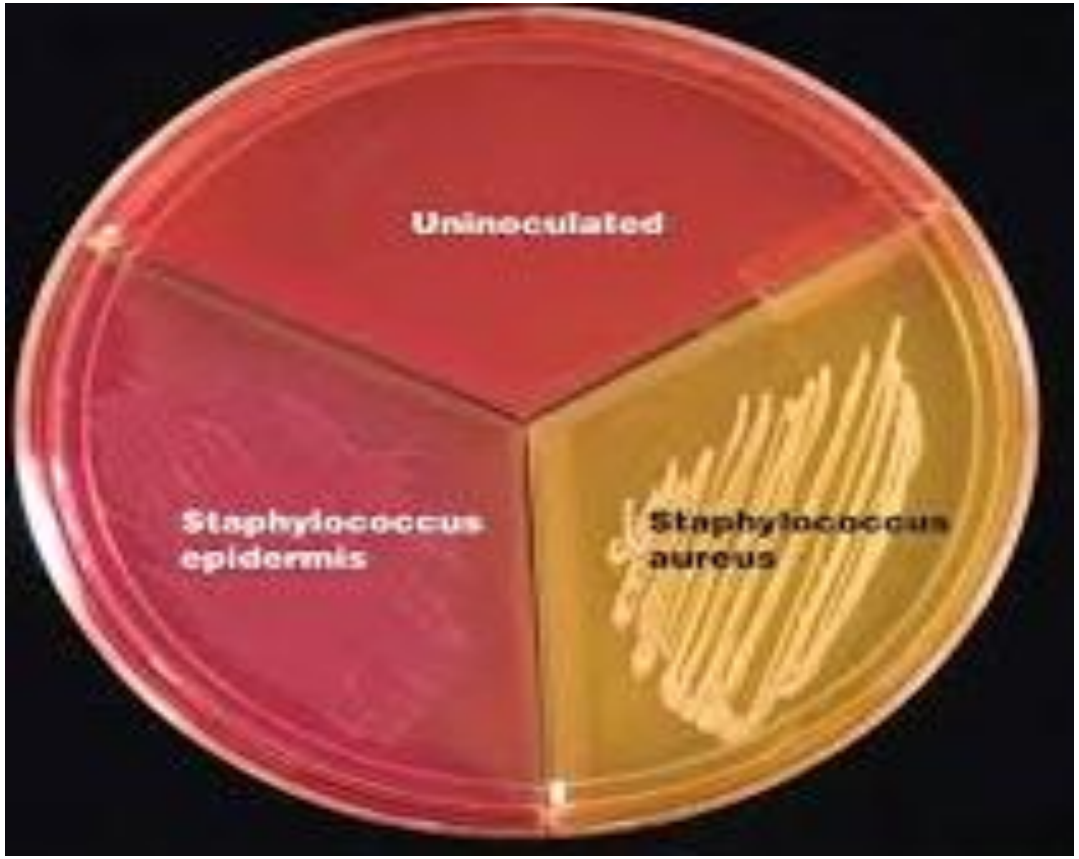
- On blood agar

-Beta hemolysis.

-20-25% Co_2.

- Selective medium:

- Mannitol fermentation on Mannitol salt agar.



- On MacConkey agar

-Pink colonies due to lactose fermentation.

- Liquid media:

-Uniform turbidity.

❖Antibiotic Prophylaxis and Therapy:

- Prescribing antibiotics in dentistry must be guided by current guidelines to minimize the selection pressure for resistant strains.
- For confirmed S. aureus infections, treatment should be based on culture and antibiotic susceptibility testing whenever possible.

❖Treatment:

- Penicillin is drug choice.
- Methicillin.
- Vancomycin.
- Superficial infection:-local application with bacitracin/ Chlorhexidine.
- Very resistant cases and chronic resistant carriers….Rifampicin along with oral antibiotic.

Prevention & Control:

1-Hand hygiene: Regular hand washing is the best preventive measure.

2-Proper wound care: Cleaning and covering wounds to prevent infection.

3-Antiseptics & Disinfection: Reducing bacterial spread in hospitals.

4-Antibiotic Stewardship: Avoiding overuse of antibiotics to prevent resistance.

Any Question?

Thank you